

of volatilities computed using short samples ending in 2007 totally missed the explosions in risks that materialized in the 2008–2009 financial crisis.

Use Economic Models

I believe that *asset allocation is fundamentally a valuation problem*. The main problem with using purely historical data, even with the profession’s best econometric toolkit, is that it usually ignores economic value. Why would you buy more of something if it is too expensive?

Valuation requires an *economic framework*. Economic models could also be combined with statistical techniques. This is the approach of Black and Litterman (1991), which is popular because it delivers estimates of expected returns that are “reasonable” in many situations. Black and Litterman start with the fact that we observe market capitalizations, or market weights. The market is a mean-variance portfolio implied by the CAPM equilibrium theory (see chapter 6). Market weights, which reflect market prices, embody the market’s expectations of future returns. Black and Litterman use a simple model—the CAPM—to reverse engineer the future expected returns (which are unobservable) from market capitalizations (which are observable). In addition, their method also allows investors to adjust these market-based weights to their own beliefs using a shrinkage estimator. I will use Black–Litterman in some examples below, in section 6.

An alternative framework for estimating inputs is to work down to the underlying determinants of value. In Part II of this book, I will build a case for thinking about the underlying factors that drive the risk and returns of assets. Understanding how the factors influence returns and finding which factor exposures are right for different investors in the long run enables us to construct more robust portfolios.

The concept of *factor investing* (see chapter 14), where we look through asset class labels to the underlying factor risks, is especially important in maximizing the benefits of diversification. Simply giving a group of investment vehicles a label, like “private equity” or “hedge funds,” does not make them asset classes. The naïve approach to mean-variance investing treats these as separate asset classes and plugs them straight into a mean-variance optimizer. Factor investing recognizes that private equity and hedge funds have many of the same factor risks as traditional asset classes. Diversification benefits can be overstated, as many investors discovered in 2008 when risky asset classes came crashing down together, if investors do not look at the underlying factor risks.

Keep It Simple (Stupid)

The simple things always work best. The main principle of mean-variance investing is to hold diversified portfolios. There are many simple diversified portfolios, and they tend to work much better than the optimized portfolios computed in the full glory of mean-variance quadratic programming in equations (3.2) and (3.3).